How the bat identify the object?

Bat emits the ultra sound frequency (more than 20 KHz) and send towards the particular object. This frequency hits the object and returns to it as echoes. The bat will calculate the location of the object by measuring the time it takes for an echo to return from it.

2.5. FUSE

A fuse is made up of a piece of metal that melts when over-heated. It should be connected in series with the load. Fuse is made up of 37% of lead and 67% of iron. Most common sizes having maximum working voltages are 32 V, 125 V, 600 V to 25000 V. Electronic fuses having a current rating of 0.1 A, 0.5 A, 1 A, 2 A, 3 A, 5 A, 10 A, 100 A and 500 A.

2.5.1 Types of fuse

Fuses can be divided into two main categories, according to the type of input supply voltage. They are,

1. AC fuses
2. DC fuses

There is a little difference between AC and DC fuses in size.

2.5.2 Other Types of Fuse

1. Cartridge Fuse
2. Blade Type Fuse
3. Surface Mount Device (SMD) Fuse
4. Axial Type Fuse
5. Thermal Type Fuse
6. HRC Type Fuse
7. Resettable Type Fuse

2.5.2.1 Cartridge Fuse

Cartridge fuses are used to protect electrical appliances such as motors, air conditioners, refrigerators, pumps etc., where high voltage rating and current rating are required. Cartridge fuse is shown in Figure 2.31



Figure 2.31 Cartridge Fuse

2.5.2.2 Blade Type Fuse

This type (also known as plug in fuses) comes in plastic body and two metal caps to fit in a socket. Mostly, they used in auto mobiles for wiring and short-circuit protection. A typical blade type fuse is shown in Figure 2.32.



Figure 2.32 Knife Blade Cartridge

2.5.2.3 Surface Mount Device (SMD) fuse

This type of fuses directly soldered in the printed circuit board. These fuses are available in low-voltage and low-current ratings. It is the latest method of fuse and also no need to pierce on the PCB. The SMD fuse is shown in Figure 2.33.

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